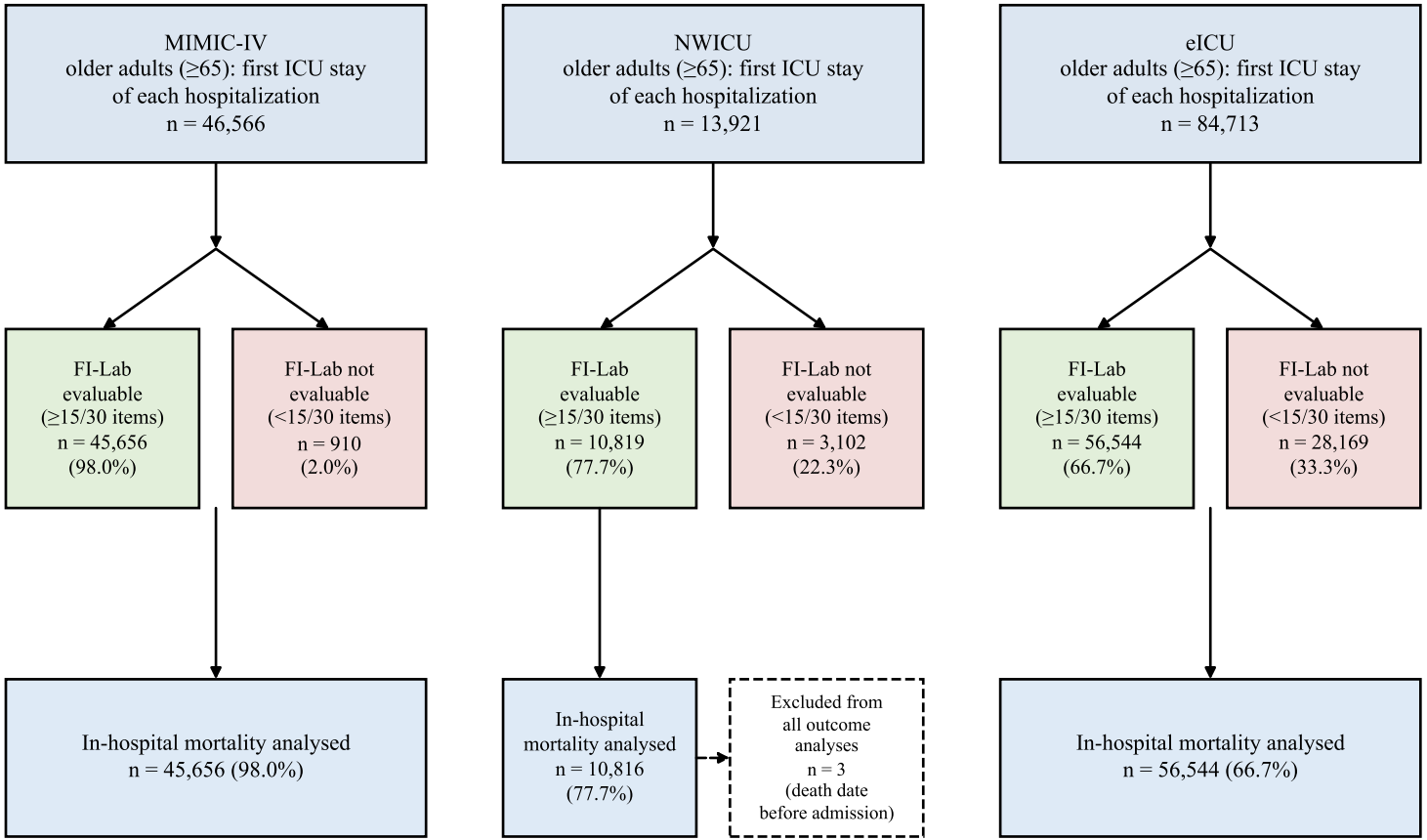
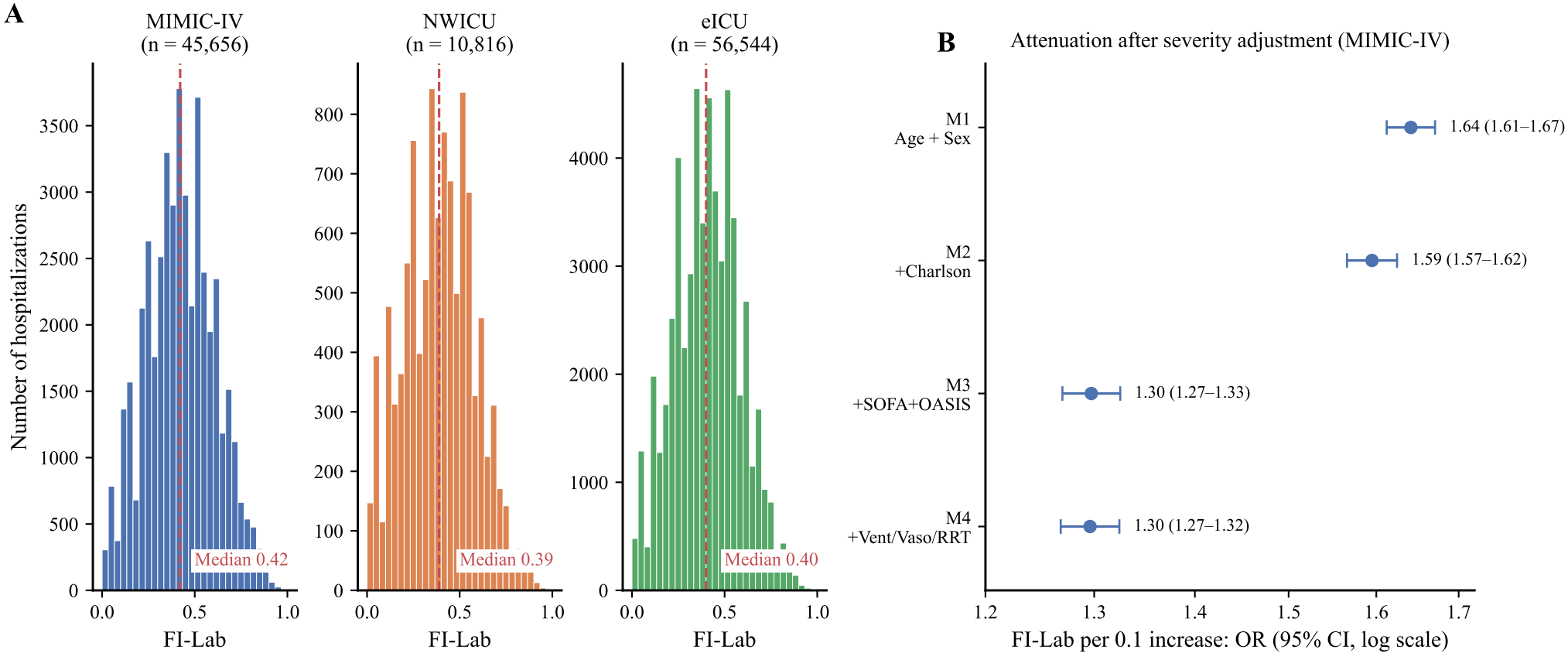
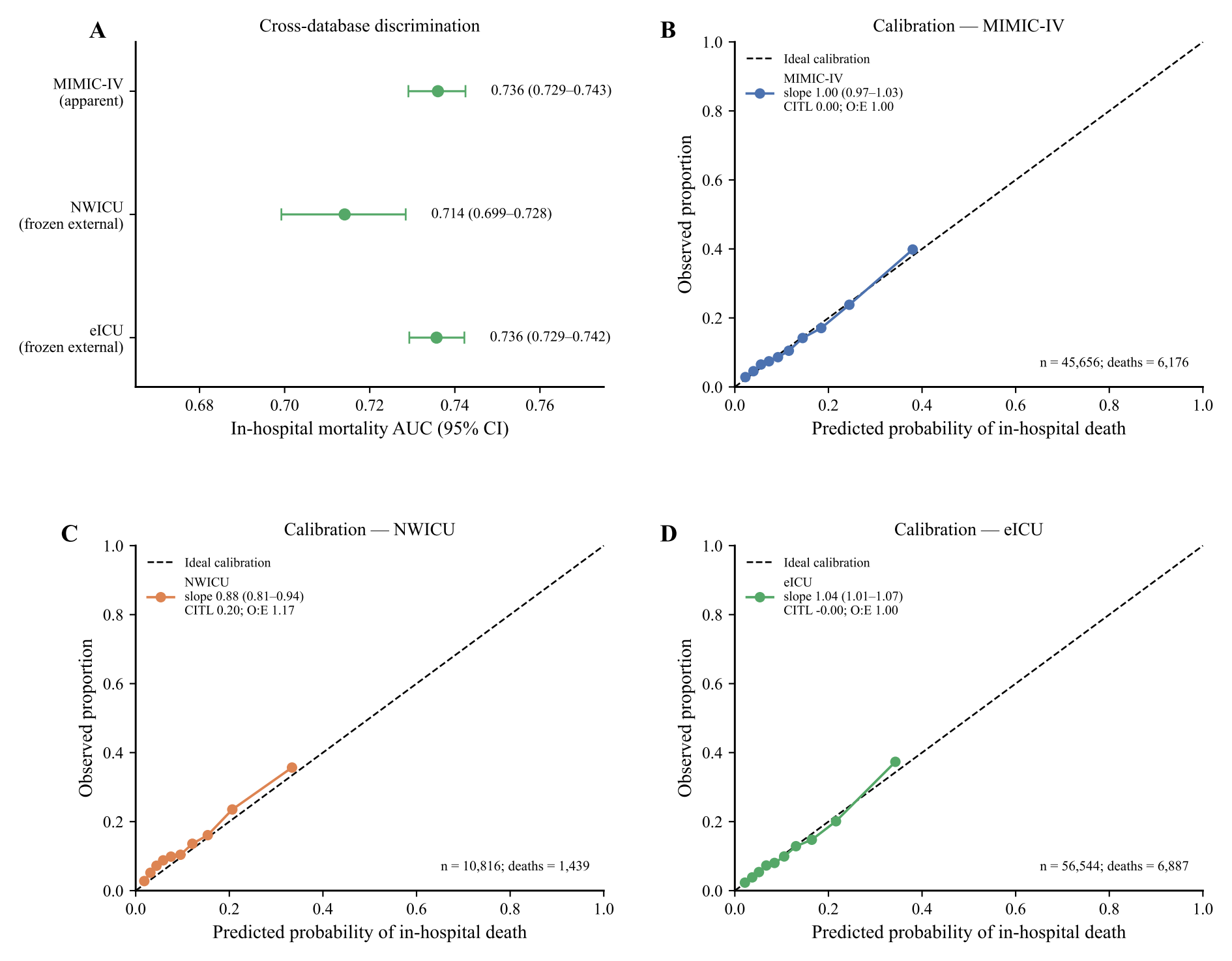


FI-Lab was computed over the first ICU day (MIMIC-IV: up to 6 h before to 24 h after ICU admission; NWICU and eICU: 0–24 h). Patients not evaluable for FI-Lab (<15 of 30 items assessed) were excluded from all analyses, so the FI-Lab–evaluable cohort is the analysis cohort of every database.
In NWICU, three hospitalizations had a recorded death date preceding the admission date; the in-hospital outcome was treated as unknown and these records were excluded from all NWICU outcome analyses (Methods 2.2). No records were excluded for this reason in MIMIC-IV or eICU.


A
MIMIC-IV
(n = 45,656)
NWICU
(n = 10,816)
eICU
(n = 56,544)
Number of hospitalizations
Median 0.42
Median 0.39
Median 0.40
FI-Lab
FI-Lab
FI-Lab
B
Attenuation after severity adjustment (MIMIC-IV)
M1
Age + Sex
1.64 (1.61–1.67)
M2
+Charlson
1.59 (1.57–1.62)
M3
+SOFA+OASIS
1.30 (1.27–1.33)
M4
+Vent/Vaso/RRT
1.30 (1.27–1.32)
FI-Lab per 0.1 increase: OR (95% CI, log scale)


A
Cross-database discrimination
MIMIC-IV (apparent)
NWICU (frozen external)
eICU (frozen external)
0.736 (0.729–0.743)
0.714 (0.699–0.728)
0.736 (0.729–0.742)
0.68
0.70
0.72
0.74
0.76
In-hospital mortality AUC (95% CI)
B
Calibration — MIMIC-IV
Ideal calibration
MIMIC-IV
slope 1.00 (0.97–1.03)
CITL 0.00; O:E 1.00
n = 45,656; deaths = 6,176
Observed proportion
Predicted probability of in-hospital death
C
Calibration — NWICU
Ideal calibration
NWICU
slope 0.88 (0.81–0.94)
CITL 0.20; O:E 1.17
n = 10,816; deaths = 1,439
Observed proportion
Predicted probability of in-hospital death
D
Calibration — eICU
Ideal calibration
eICU
slope 1.04 (1.01–1.07)
CITL -0.00; O:E 1.00
n = 56,544; deaths = 6,887
Observed proportion
Predicted probability of in-hospital death